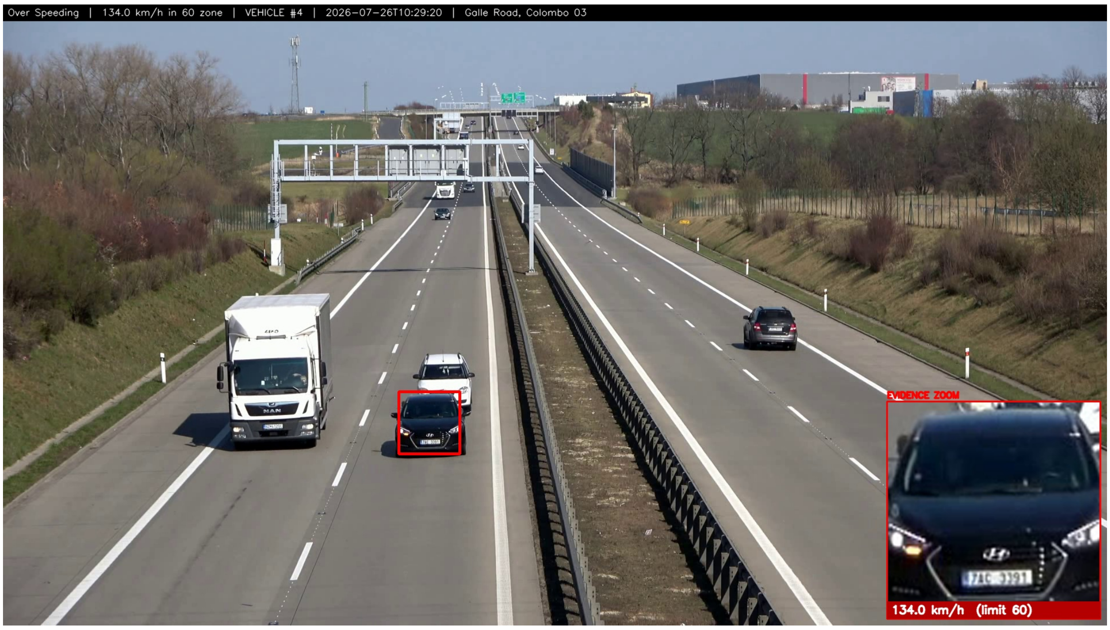
AI Smart Traffic Intelligence Platform

Technical Documentation, Models and Verified Results



System output: over-speeding evidence, 134.0 km/h in a 60 km/h zone, with number-plate zoom inset.

All figures were measured on the development machine (12-core CPU, no GPU). Nothing is estimated.

1. Overview

The platform turns an ordinary camera - fixed CCTV, a phone stream or a drone - into an automated traffic officer. It detects and tracks every vehicle, judges nine categories of violation, reads number plates, measures true road speed, and issues an e-challan carrying photographic evidence to the police. It runs on a laptop CPU.

2. Processing pipeline

Each analysed frame passes through seven stages. Stages 3-5 are what separate this from a vehicle-counting demo.

#	Stage	What happens
1	Capture	Frame pulled from CCTV / file / webcam; oversized frames downscaled
2	Detection	YOLOv8s finds vehicles, people, traffic lights and phones
3	Tracking	ByteTrack assigns a persistent ID so a vehicle is followed across frames
4	Geometry	Homography maps image pixels onto real road metres
5	Rules	Violation engine applies nine rules behind false-positive gates
6	ANPR	Plate localised, cropped, OCR-read and confirmed by repeated agreement
7	Evidence	Annotated snapshot, challan record, PDF and police email

3. Models used and why

Model	Role	Why this one
YOLOv8s	Vehicles, people, lights, phones	Upgraded from YOLOv8n after measurement: on identical footage yolov8n found 64 motorcycles and cou
ByteTrack	Multi-object tracking	Associates low-confidence detections too, so a vehicle keeps its ID through partial occlusion. Persistent I
EasyOCR	Plate text	Reads Latin-script plates without per-country training, matching Sri Lankan plates. Never reformats to a r
license_plate_ detector.pt	Plate localisation	Two-stage ANPR: find the plate, then OCR only that crop. Far more accurate than OCR-ing a whole vehi
helmet.pt	Helmet / no helmet	COCO has no helmet class. Applied to each associated rider's own head region, not the bike area.
yolov11s-seatbelt	Seatbelt	Drop-in model with no_seatbelt / seat_belt classes. Without it the rule stays off - there is no guess fallbac

4. Core algorithms

4.1 Speed by perspective transform

Four surveyed road corners are mapped to a real-world rectangle using a homography (cv2.getPerspectiveTransform). Each vehicle's ground-contact point is projected into road metres, and speed is a least-squares regression of metres against time across a 2.5-second window - never a single-frame difference. Outliers beyond two standard deviations are dropped, the result is smoothed, and a reading is only trusted when the fit is straight (R-squared above 0.55), the residual is under 2.5 m and the track has been watched for at least one second.

4.2 Motion gate

A violation requires genuine movement. Net displacement is measured over a time-based window with camera-motion compensation, so box jitter on a parked vehicle cancels out. This single gate is why parked motorcycles are never fined.

4.3 Multi-frame persistence

Every appearance rule must be observed repeatedly before firing - three frames for helmet and triple riding, four for phone use. A one-frame anomaly, the classic demo-killer, can never issue a challan.

4.4 Plate confirmation by voting

OCR reads are pooled by their digit tail, so garbled letters still vote together. A plate is confirmed only when three separate reads agree. Measured effect: two-read confirmation produced EEBBJ 8752 for a plate that actually reads NP BBJ 8752; requiring three recovered BBJ 8752.

4.5 Real-time frame dropping

CPU detection is slower than playback, so the analyser drops frames it was too busy for - exactly as a live camera does - and the video plays at true speed. Measured at 0.99-1.01x. For a live camera, capture runs on its own thread and the last annotation is stamped onto fresh frames, so the picture stays smooth while boxes refresh behind it.

5. The nine rules

Rule	Judged by	Precondition
No Helmet	Helmet model on each rider's head region	Motorcycle + rider
Triple Riding	Riders associated with one motorcycle	3 riders detected
Over Speeding	Regression of road-metres against time	Camera calibration
Red Light Jump	Stop-line crossing while the governing signal is red	Signal ROI + lane zone
Wrong Way	Travel direction against the policed lane	Direction + lane zone
No Seatbelt	Seatbelt model on the windscreen region	models/seatbelt.pt
Mobile Phone Use	Phone attributed to a rider or driver	Visible phone
Illegal Parking	Stationary past a threshold in a no-parking zone	Fixed camera
Driver Fatigue	Continuous driving without a qualifying break	Long observation

6. Measured results

Test	Recorded result
Automated test suite	79 tests passing across 7 suites
sample_1080p.mp4 (calibrated)	7 Over Speeding events, speeds 83-163 km/h
stream.mp4 (calibrated)	3 Over Speeding, 11 of 52 vehicles measured, max 106 km/h
IMG_6992.MOV (Sri Lanka)	150 vehicles; plates AAG 4002 and BBJ 8752 read
srilanka.mp4 (parked bikes)	41 vehicles, 0 violations - a correct refusal
Single-image upload	No Helmet + Triple Riding, 2 challans issued
Real-time replay	0.99-1.01x true speed via frame dropping
Live webcam	1.89 to ~7 published fps after decoupling capture
Warm image analysis	~2 seconds end to end

7. Defects found by testing on real footage

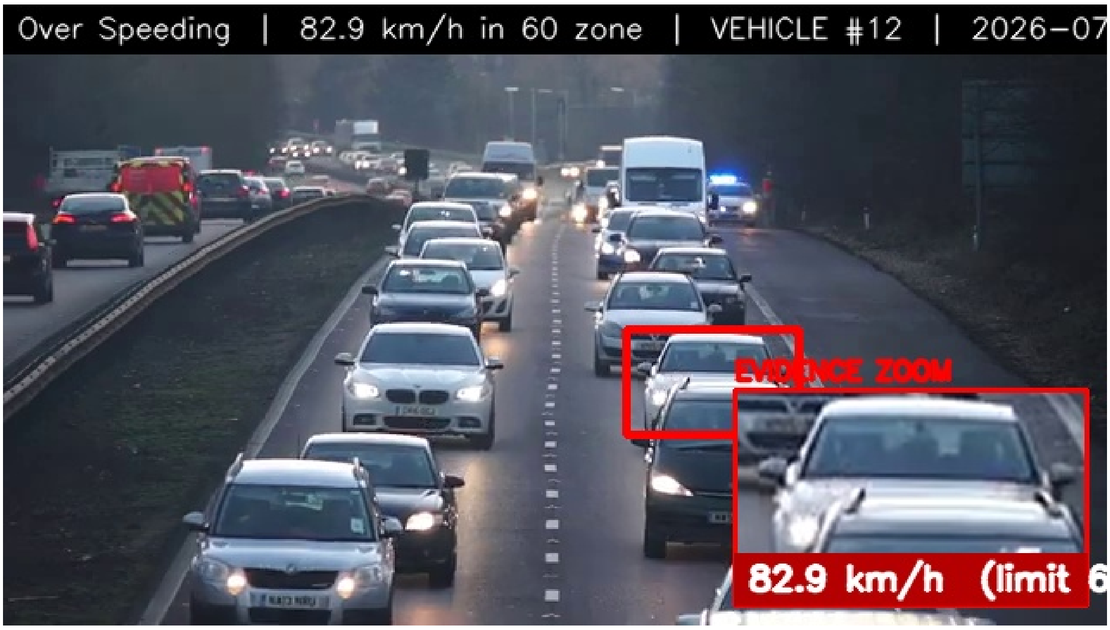
A traffic system that issues wrong fines is worse than no system. Each defect below was found by running against real video, and each is now covered by the automated suite.

Defect	Cause	Fix
Moving vehicles fined for parking	Motion gate needed 3 samples in a 1-second frame window; at ~2 fps, 89% of moving vehicles read as stationa	Time-based window; a track ever seen moving is exempt
Helmeted riders accused	Helmet model ran on a crop wider than the bike, catching neighbours	Judge each rider's own head region
Bare heads marked compliant	Compliance needed 0.35 confidence against 0.50 to accuse	Decide by strongest signal; raise the bar to 0.55
Stopped car fined for red light	Any light in frame applied to any vehicle, but junctions have several heads facing different approaches	SIGNAL_ROI selects the governing head; RED_LIGHT_ZONE limits th
Wrong plate on challans	Two agreeing OCR reads let two wrong reads agree	Require three agreeing reads
Triple riding never fired	Occluded pillion riders never cleared the 0.40 person threshold	Lower bar (0.25) for rider association only

8. Verified system output

Single-image analysis: three riders and no helmets detected on a CC-licensed public photograph, producing two separate challans.

Live Sri Lankan road: 18 vehicles tracked, plate AAG 4002 read and confirmed, speed shown beside each vehicle ID.



Highway over-speeding evidence generated from a second calibrated camera, showing the method is not tied to one clip.

9. Use cases

Deployment	What it delivers
Urban junction enforcement	Red-light, helmet, triple-riding and phone-use challans on existing CCTV, with evidence strong enough to
Highway speed corridor	Calibrated speed enforcement without physical speed guns or officer presence, running continuously
School and hospital zones	No-parking zone monitoring plus low speed limits, where illegal stopping directly endangers pedestrians
Commercial fleet oversight	Driver-fatigue and seatbelt monitoring for buses and lorries, supporting transport-authority driving-hour ru
Drone patrol	Temporary enforcement at events, accident sites or roads without fixed cameras
Road-safety analytics	Searchable plate log, violation hotspots and repeat-offender identification to target scarce police resource

10. Data handling and ethics

Number plates identify real people, so the system is deliberately conservative. A plate is written only after three independent reads agree; anything less is stored as UNREADABLE and marked for manual review. Evidence images and plate crops stay on the operator's machine and are attached only to the police challan for that specific violation. The system records what it observed and never infers identity, intent or history beyond the plate itself.

11. Known limits

Stated plainly, because anyone deploying this must understand them:

- Speed requires per-camera calibration; without it no speed is shown.
- Plate OCR needs roughly 40 or more pixels of plate width.
- Red-light enforcement requires knowing which signal head governs the policed lane.
- Illegal parking requires a fixed camera - a moving camera cannot establish that a vehicle is stationary.
- About 2 frames per second are analysed on a CPU laptop; the video stays real-time by dropping frames.
- Detection degrades at night and in heavy rain, as with any camera system.

12. Roadmap

Night and adverse-weather models; a vision-language model such as Moondream to resolve cases the detector currently reports as not judged; a trained three-wheeler class, since COCO has none and Sri Lankan tuk-tuks are currently labelled as trucks; GPU edge deployment at a live junction; and a city-wide dashboard aggregating multiple cameras.